

Mateus F. Paiva

Publications and Conference Contributions

Peer-Reviewed Journal Articles

1. **Synthesis and Properties of BEA Zeolite Modified with Iron(III) Oxide.** *Inorganics* **13**, 383 (2025).
<https://doi.org/10.3390/inorganics13120383>
2. **Advancing Lignocellulosic Biomass Fractionation through Molten Salt Hydrates: Catalyst-Enhanced Pretreatment for Sustainable Biorefineries.** *ChemSusChem* **17**, e202400396 (2024).
<https://doi.org/10.1002/cssc.202400396>
3. **Cellulose Depolymerization Using Zinc Chloride Hydrate and Solid Acid Catalysts.** *Cellulose* **31**, 7953–7972 (2024).
<https://doi.org/10.1007/s10570-024-06107-0>
4. **Polymers Based on PLA from Synthesis Using D,L-Lactic Acid and Some Biomedical Applications: A Short Review.** *Polymers* **14**, 2317 (2022).
<https://doi.org/10.3390/polym14122317>
5. **Structural and Acidity Analysis of Heteropolyacids Supported on Faujasite Zeolite and Its Effect in Esterification.** *Molecular Catalysis* **532**, 112737 (2022).
<https://doi.org/10.1016/j.mcat.2022.112737>
6. **Synthesis, Characterization, and Application of Phosphotungstic Acid Supported on Iron-Based Magnetic Nanoparticles.** *Catalysis Today* **394–396**, 425–433 (2021).
<https://doi.org/10.1016/j.cattod.2021.06.032>
7. **Dehydration of Fructose to 5-Hydroxymethylfurfural over Solid Acid Catalysts.** *Chemistry* **3**, 1189–1202 (2021).
<https://doi.org/10.3390/chemistry3040087>
8. **Accessibility and Strength of H-Acceptor Hydroxyls of Ordered Mesoporous Silicas.** *Journal of Porous Materials* **28**, 323–335 (2020).
<https://doi.org/10.1007/s10934-020-00994-x>
9. **Preparation of PLA Blends by Polycondensation of D,L-Lactic Acid Using Supported Heteropolyacids.** *Heliyon* **5**, e01810 (2019).
<https://doi.org/10.1016/j.heliyon.2019.e01810>
10. **Comparative Acidity of BEA and Y Zeolite Composites with Heteropolyacids.** *Molecular Catalysis* **458**, 152–160 (2018).
<https://doi.org/10.1016/j.mcat.2018.03.005>
11. **Acidity and Characterization of 12-Tungstophosphoric Acid Supported on Silica-Alumina.** *J. Braz. Chem. Soc.* **28**, 336–347 (2017).
<https://doi.org/10.5935/0103-5053.20160183>
12. **Generation and Characterization of Catalytically Active Sites of Heteropolyacids on Zeolite Y.** *Catalysis Today* **289**, 70–77 (2017).
<https://doi.org/10.1016/j.cattod.2016.08.010>

Book Chapters

13. **A Review of the Synthesis of Cerium Oxide: Effects on Structure and Oxygen Storage Capacity.** In: *Cerium Oxide: Structure, Occurrence and Applications*. Nova Science Publishers, New York, 53–92 (2022).
 14. **Catalysis of Fructose Conversion to HMF via Inorganic Solid Acids.** In: *Chemistry: Debate between Modern Life and the Environment 2*. Atena Editora, 177–185 (2021).
<https://doi.org/10.22533/at.ed.79021080418>
 15. **Synthesis, Characterization and Properties of Molecular Sieves: SBA-16, FDU-12, SBA-15 and MCM-41.** In: *Exact and Earth Sciences: Strategic Knowledge for National Development*. Atena Editora, 146–159 (2020).
<https://doi.org/10.22533/at.ed.60220020712>
 16. **Incipient Wetness Impregnation of HSiW on Zeolite Y for Butyl Acetate Production.** In: *Interdisciplinary Studies in Exact, Earth Sciences and Engineering 5*. Atena Editora, 287–297 (2019).
<https://doi.org/10.22533/at.ed.23219110926>
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Conference Contributions – Full Papers in Proceedings

17. **Selective Production of Monophenols and Alkylphenols in Organosolv Lignin Depolymerization.** *23rd Brazilian Congress on Catalysis (CBCat)*, Natal (2025).
18. **Lignocellulosic Biomass Fractionation Using Molten Salt Hydrates and Catalytic Lignin Depolymerization.** *23rd Brazilian Congress on Catalysis (CBCat)*, Natal (2025).
19. **Calorimetric and Structural Analysis of Faujasite-Supported Catalysts in Oleic Acid Esterification.** *21st Brazilian Congress on Catalysis*, Rio de Janeiro (2021).
20. **Effect of Heteropolyacid Incorporation Route on Zeolite Y-Based Catalysts.** *XXVI Ibero-American Congress on Catalysis (CiCat)*, Coimbra (2018).
21. **Synthesis, Characterization and Acidity of Ordered Mesoporous Silicas.** *XXVI Ibero-American Congress on Catalysis (CiCat)*, Coimbra (2018).
22. **Polycondensation of D,L-Lactic Acid for PLA Production Using Heterogeneous Catalysts.** *XII Catalysis Meeting – N/NE/CO Regions*, Belém (2018).
23. **Supported 12-Tungstophosphoric Acid Catalysts for Lactic Acid Polymerization.** *IX Mercosur Congress on Catalysis / 19th CBCat*, Ouro Preto (2017).
24. **Synthesis of H3PW12O40 Catalysts on Y-Type Zeolites.** *69th Annual Meeting of SBPC*, Belo Horizonte (2017).
25. **Influence of Impregnation Methodology on HPW/HSiW Zeolite Y Catalysts.** *18th Brazilian Congress on Catalysis*, Porto Seguro (2015).
26. **Synthesis and Characterization of HPW/HSiW Catalysts on BEA Zeolite.** *55th Brazilian Congress of Chemistry*, Goiânia (2015).

Conference Contributions – Extended Abstracts and Abstracts

27. **Exploring n-Propylbenzene Ignition for Sustainable Aviation Fuels.** *Low-Carbon Combustion Meeting*, Nancy (2024).
28. **Comparative Acidity of Zeolite Y and BEA Composites with Heteropolyacids.** *8th International Symposium on Acid-Base Catalysis*, Rio de Janeiro (2017).
29. **Acidic Properties of Supported Heteropolyacids on Zeolite Y.** *18th International Zeolite Conference (IZC)*, Rio de Janeiro (2016).
30. **Selective Hydrolysis of Hemicellulose Using $\text{LiBr} \cdot 3\text{H}_2\text{O}$.** *48th Annual Meeting of SBQ*, Campinas (2025).
31. **Reductive Catalytic Depolymerization of Organosolv Lignin over $\text{Pd/Nb}_2\text{O}_5$.** *16th European Congress on Catalysis (EuropaCat)*, Trondheim (2025).
32. **Fractionation and Depolymerization of Eucalyptus in Lithium Bromide Solution.** *18th International Congress on Catalysis (ICC)*, Lyon (2024).
33. **Titania-Supported Palladium Catalyst for Lignin Reductive Fractionation.** *Congrès de la Société Chimique de France (SCF)*, Nantes (2023).
34. **Catalytic Depolymerization of Cellulose over ZSM-5 in Molten Salt Media.** *28th North American Catalysis Society Meeting*, Providence (2023).
35. **Silica-Alumina as Catalyst for Melt Polycondensation of L-Lactic Acid.** *27th North American Catalysis Society Meeting*, New York (2022).
36. **Cellulose Upgrading Using ZnCl_2 Molten Salt and Solid Acid Oxides.** *27th North American Catalysis Society Meeting*, New York (2022).
37. **Properties of PLA Polymers Produced by Heterogeneous Catalytic Polycondensation.** *42nd Annual Meeting of SBQ*, Joinville (2019).
38. **Preparation of Poly(lactic acid) Blends by Heterogeneous Catalysis.** *41st Annual Meeting of SBQ*, Foz do Iguaçu (2018).
39. **Catalytic Synthesis of Butyl Oleate by Supported Heteropolyacids.** *46th World Chemistry Congress / 40th SBQ Meeting*, São Paulo (2017).
40. **Structural Evaluation of Zeolite Y as Support for Keggin Heteropolyacids.** *10th Catalysis Regional Meeting (N/NE/DF)*, Recife (2014).